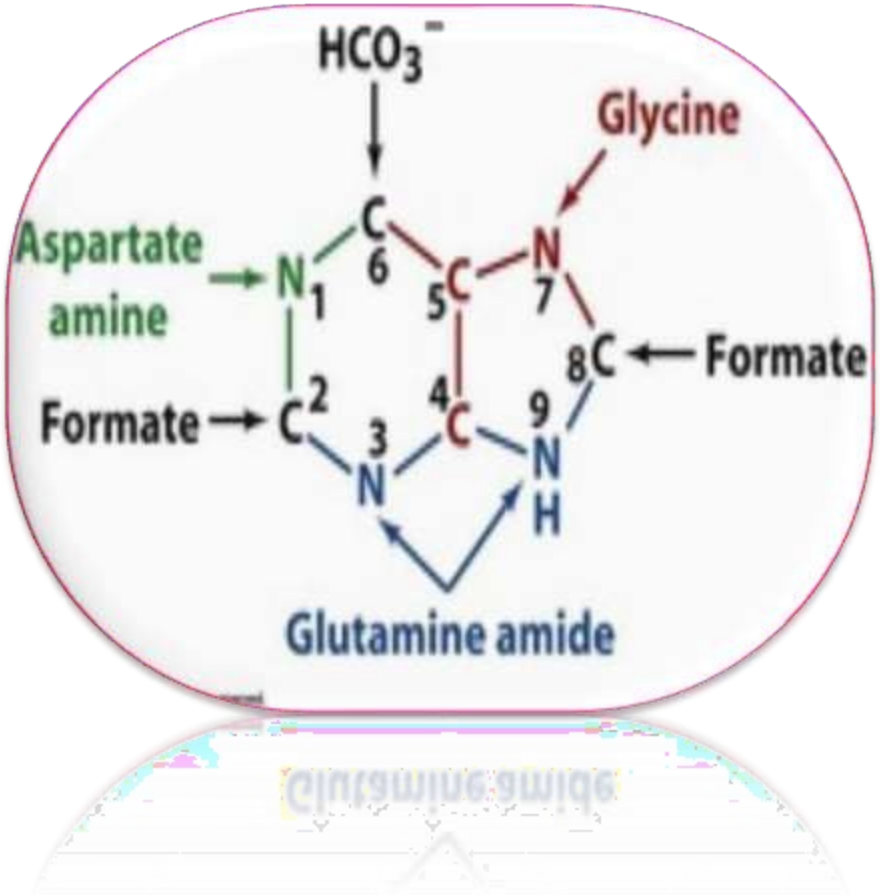
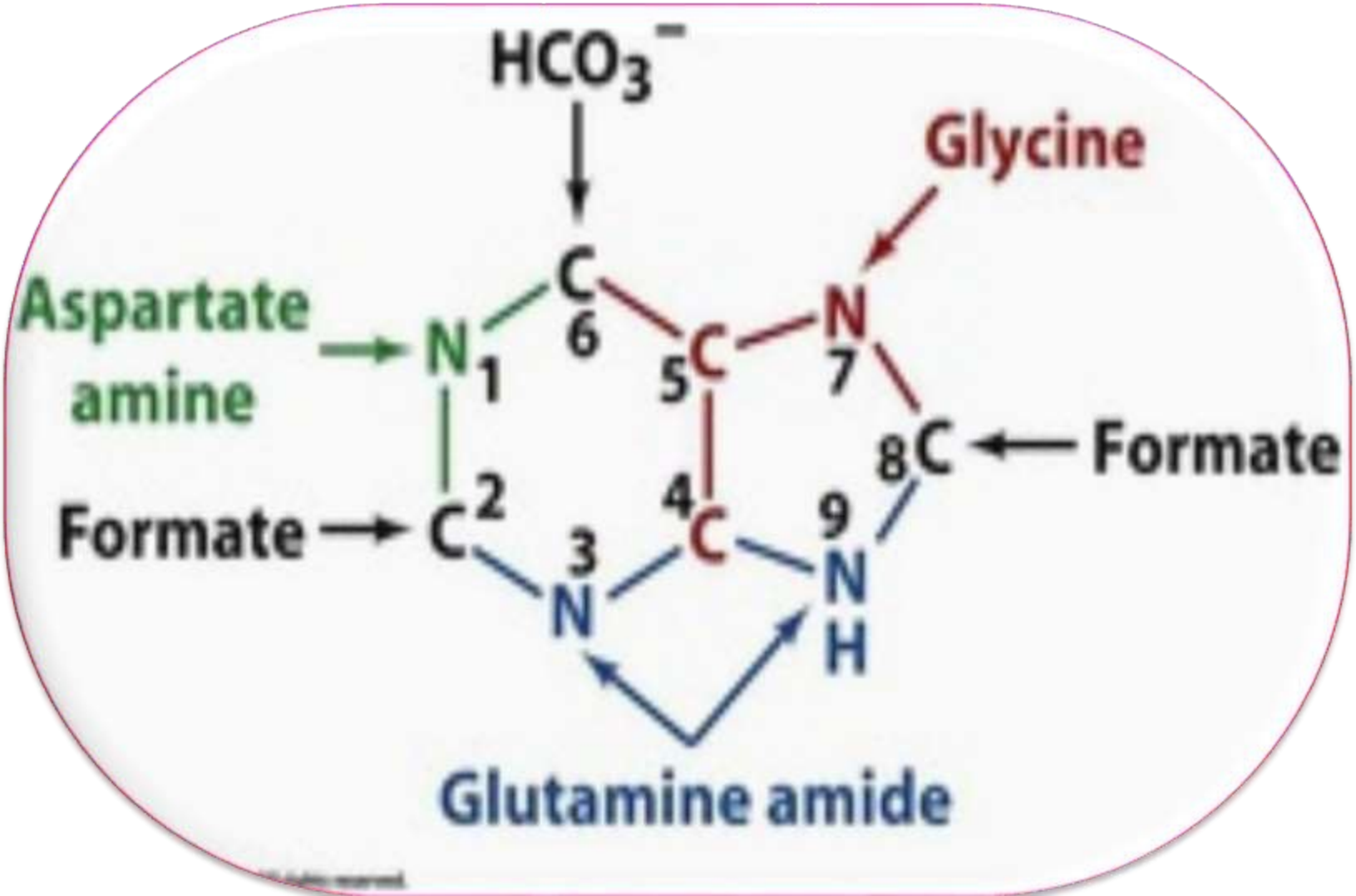
Biosynthesis of Purine Ribonucleotides



Formation of purine ring

- N1 of purine is derived from amino group of aspartate.
- C2 & C8 from formate of N^{10} - formyl THF.
- N3 & N9 are obtained from amide group of glutamine.
- C4, C5 & N7 are contributed by glycine.
- C6 directly comes from CO_2.

Formation of purine ring



Glutamine amide

- **The purines are built upon a pre-existing ribose 5-phosphate.**
- **Liver is the major site for purine nucleotide synthesis.**
- **Erythrocytes, polymorphonuclear leukocytes & brain cannot produce purines.**

Steps in Purine Biosynthesis

- **Ribose 5-phosphate, of carbohydrate metabolism is the starting material for purine nucleotide synthesis.**
- **It reacts with ATP to form phosphoribosyl pyrophosphate (PRPP).**
- **Glutamine transfers its amide nitrogen to PRPP to replace pyrophosphate & produce 5-phosphoribosylamine.**

- PRPP glutamyl amidotransferase is controlled by feedback inhibition of nucleotides (IMP, AMP & GMP).
- This reaction is the 'committed.
- Phosphoribosylamine reacts with glycine in the presence of ATP to form glycinamide ribosyl 5-phosphate or glycinamide ribotide (GAR).
- Catalyzed by synthetase.

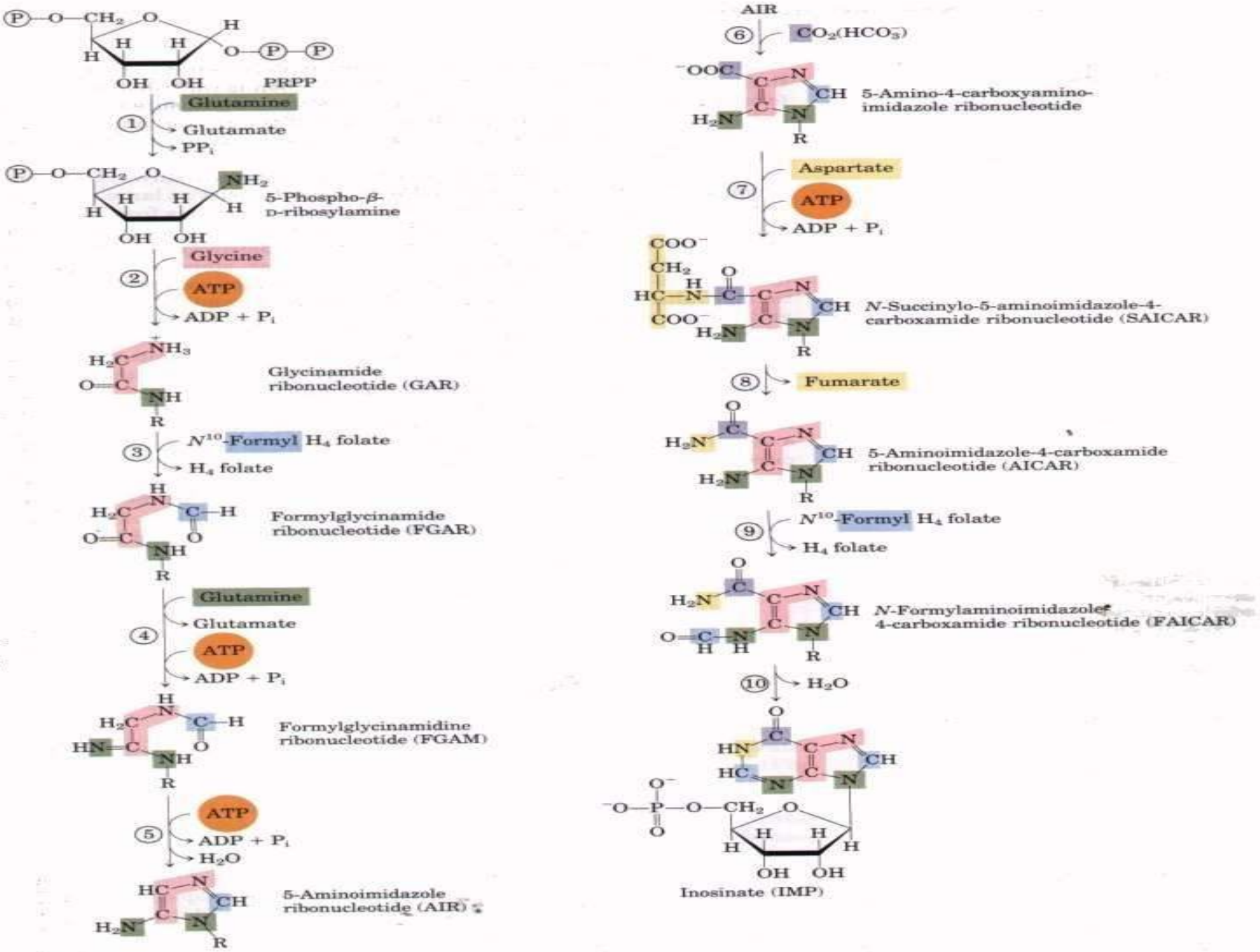
- N^{10}-Formyl tetrahydrofolate donates the formyl group & the product formed is formylglycinamide ribosyl 5-phosphate.
- The reaction is catalyzed by formyltransferase.
- Glutamine transfers the second amido amino group to produce formylglycinamidine ribosyl 5-phosphate.
- The reaction is catalyzed by synthetase.

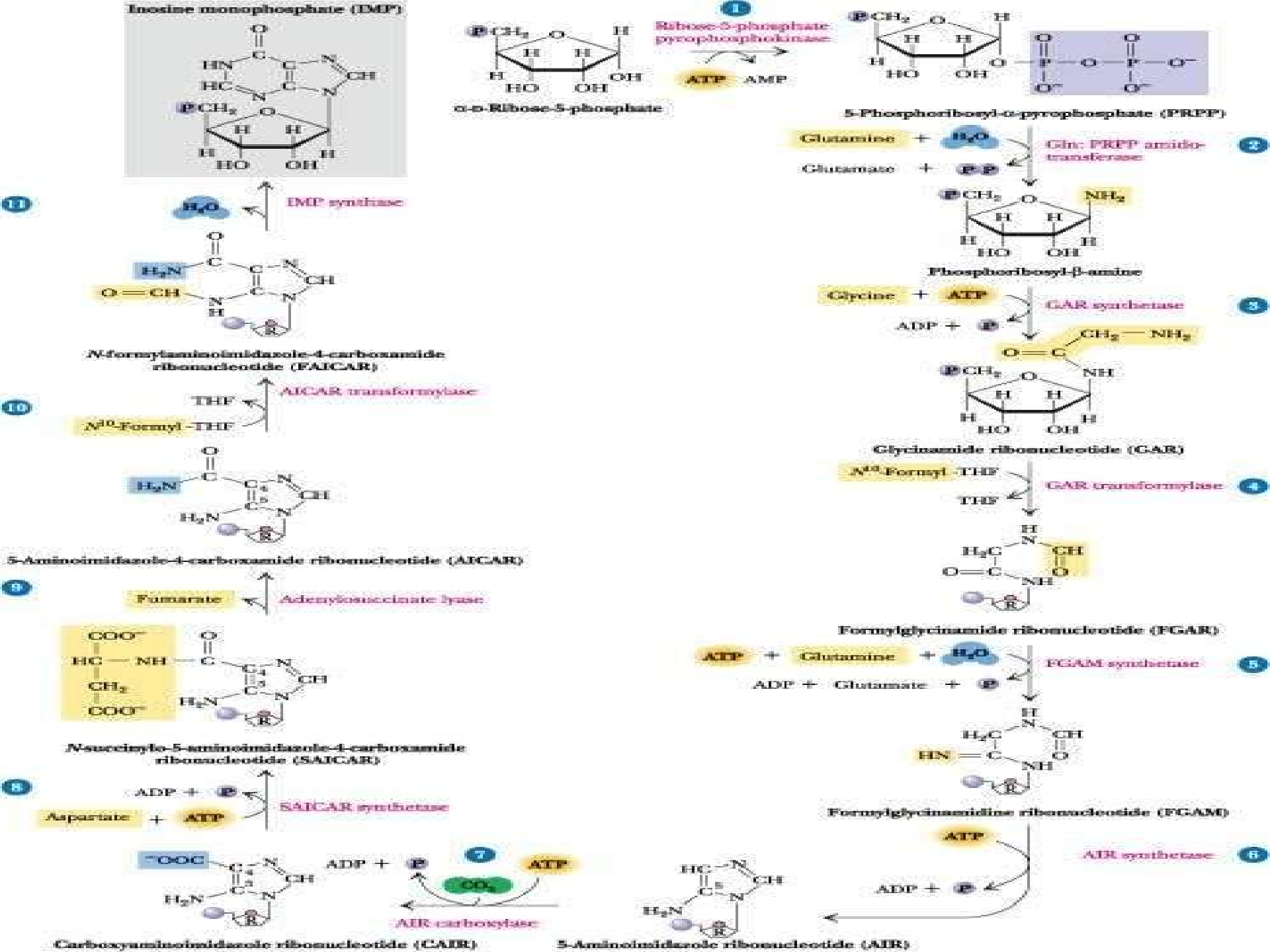
- The imidazole ring of the purine is closed in an ATP dependent reaction to yield 5-aminoimidazole ribosyl 5-phosphate.
- The reaction is catalyzed by synthetase.
- Incorporation of CO_2 (carboxylation) occurs to yield aminoimidazole carboxylate ribosyl 5-phosphate.
- The reaction is catalyzed by carboxylase.

- Does not require the vitamin biotin or ATP.
- Aspartate condenses with the aminoimidazole carboxylate ribosyl 5-phosphate to form aminoimidazole 4-succinylcarboxamide ribosyl 5-phosphate.
- The reaction is catalyzed by synthetase.

- **Adenosuccinate lyase cleaves off fumarate & only the amino group of aspartate is retained to yield aminoimidazole 4-carboxamide ribosyl 5-phosphate.**
- **N^{10}-Formyl tetrahydrofolate donates a one-carbon moiety to produce 5-formaminoimidazole 4-carboxamide ribosyl 5-phosphate.**
- **Catalyzed by formyltransferase.**

- The final reaction catalyzed by cyclohydrolase leads to ring closure with an elimination of water molecule.
- The product obtained is inosine monophosphate (IMP), the parent purine nucleotide from which other purine nucleotides can be synthesized.


PRPP
Glutamine
Glutamate
PP_i
5-Phospho-β-D-ribosylamine
Glycine
ATP
ADP + P_i
Glycinamide ribonucleotide (GAR)
N^{10}-Formyl H_4 folate
H_4 folate
Formylglycinamide ribonucleotide (FGAR)
Glutamine
Glutamate
ATP
ADP + P_i
Formylglycinamidine ribonucleotide (FGAM)
ATP
ADP + P_i
H_2O
5-Aminoimidazole ribonucleotide (AIR)
AIR
$CO_2(HCO_3^-)$
5-Amino-4-carboxyamino-imidazole ribonucleotide
Aspartate
ATP
ADP + P_i
N-Succinylo-5-aminoimidazole-4-carboxamide ribonucleotide (SAICAR)
Fumarate
5-Aminoimidazole-4-carboxamide ribonucleotide (AICAR)
N^{10}-Formyl H_4 folate
H_4 folate
N-Formylaminoimidazole-4-carboxamide ribonucleotide (FAICAR)
H_2O
Inosinate (IMP)

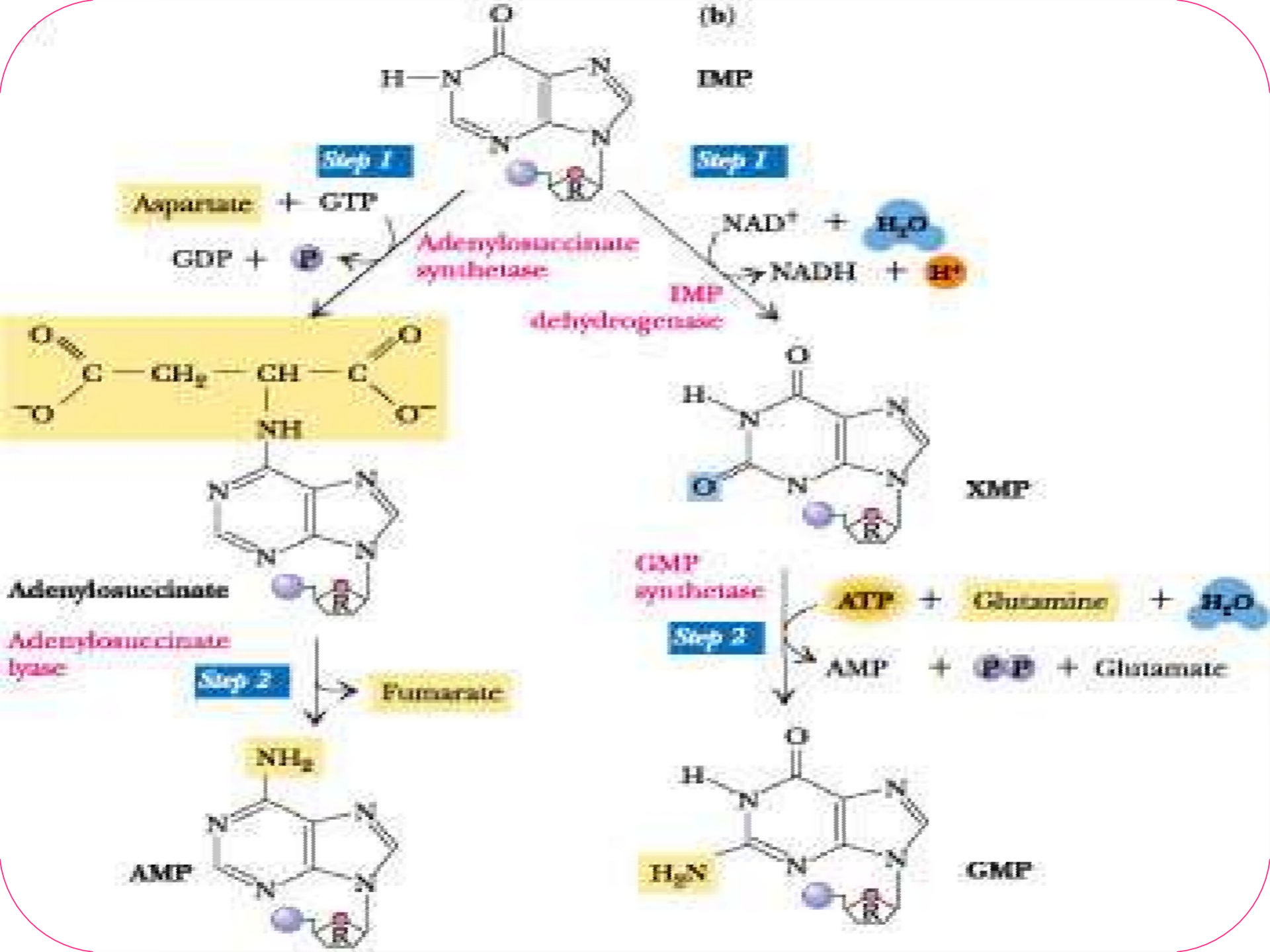

Inosine monophosphate (IMP)
1
Ribose-5-phosphate pyrophosphokinase
ATP
AMP
α-D-Ribose-5-phosphate
5-Phosphoribosyl-α-pyrophosphate (PRPP)
2
Glutamine + H2O
Glutamate + PP
Gln: PRPP amido-transferase
Phosphoribosyl-β-amine
3
Glycine + ATP
ADP + P
GAR synthetase
Glycinamide ribonucleotide (GAR)
4
N10-Formyl-THF
THF
GAR transformylase
Formylglycinamide ribonucleotide (FGAR)
5
ATP + Glutamine + H2O
ADP + Glutamate + P
FGAM synthetase
Formylglycinamidine ribonucleotide (FGAM)
6
ATP
ADP + P
AIR synthetase
5-Aminoimidazole ribonucleotide (AIR)
7
ATP
ADP + P
CO2
AIR carboxylase
Carboxyaminoimidazole ribonucleotide (CAIR)
8
Aspartate + ATP
ADP + P
SAICAR synthetase
N-succinylo-5-aminoimidazole-4-carboxamide ribonucleotide (SAICAR)
9
Fumarate
Adenylosuccinate lyase
5-Aminoimidazole-4-carboxamide ribonucleotide (AICAR)
10
THF
N10-Formyl-THF
AICAR transformylase
N-formylaminoimidazole-4-carboxamide ribonucleotide (FAICAR)
11
H2O
IMP synthase

Synthesis of AMP & GMP from IMP

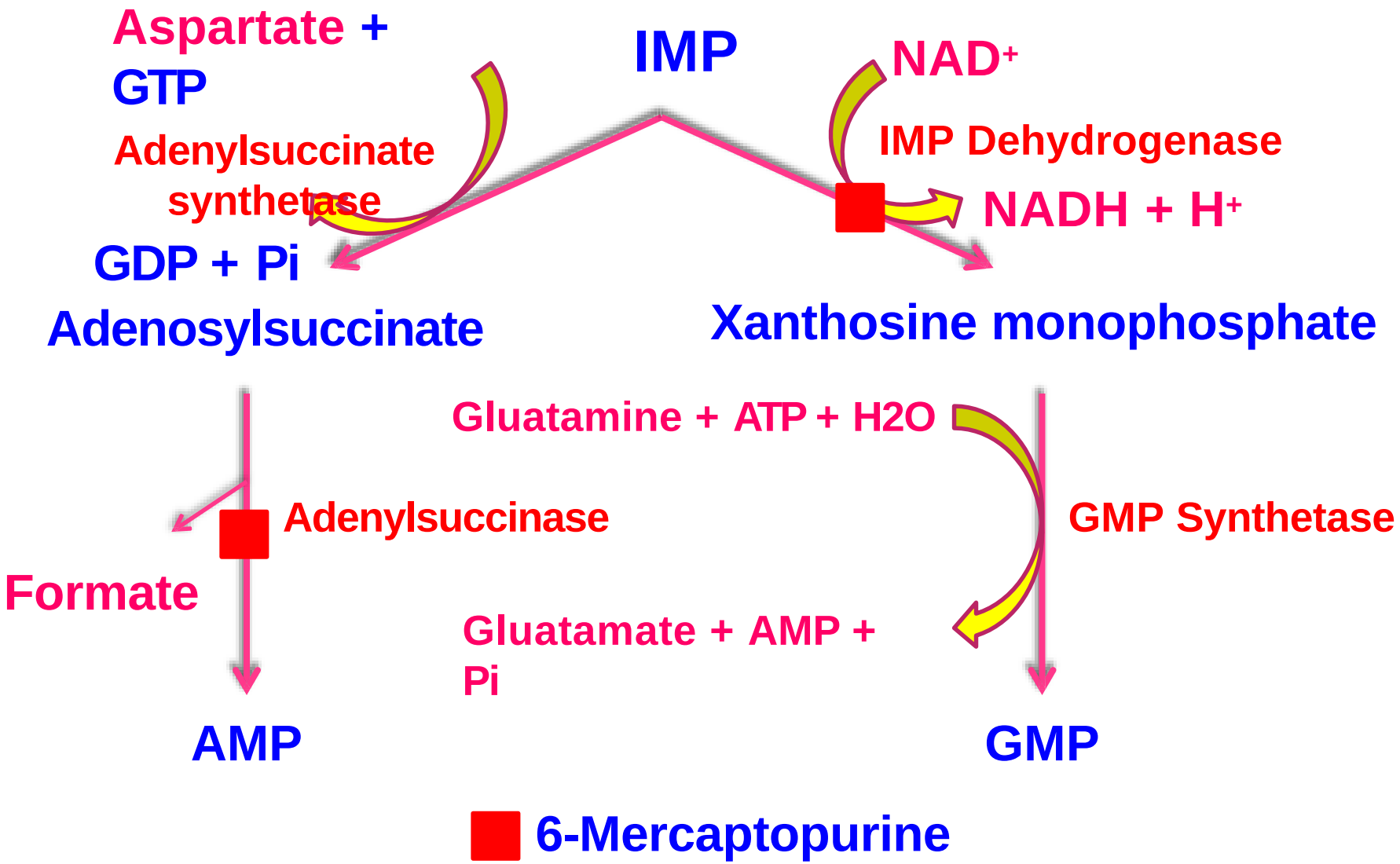
- **Synthesis of AMP:**
- **Ionosine monophosphate (IMP) is the immediate precursor for the formation of AMP & GMP.**
- **Aspartate condenses with IMP in the presence of GTP to produce adenylsuccinate which, on cleavage, forms AMP.**

- **Synthesis of GMP:**
- **IMP undergoes NAD^+ dependent dehydrogenation to form xanthosine monophosphate (XMP).**
- **Glutamine then transfers amide nitrogen to xanthosine monophosphate (XMP) to produce GMP.**

- **6-Mercaptopurine is an inhibitor of the synthesis of AMP & GMP.**
- **It acts on the enzyme adenylsuccinase (of AMP pathway).**
- **IMP dehydrogenase (of GMP pathway).**


(b)
IMP
Step 1
Aspartate + GTP
GDP + P
Adenylosuccinate synthetase
Step 1
NAD+ + H2O
NADH + H+
IMP dehydrogenase
NH
Adenylosuccinate
XMP
GMP synthetase
ATP + Glutamine + H2O
Step 2
AMP + PP + Glutamate
Adenylosuccinate lyase
Step 2
Fumarate
NH2
AMP
H2N
GMP

Synthesis of AMP & GMP



Formation of di & tri-phosphates

- The nucleoside monophosphates (AMP & GMP) converted to the corresponding di & triphosphates.
- This is achieved by the transfer of phosphate group from ATP, catalysed by nucleoside monophosphate (NMP) kinases & nucleoside diphosphate (NDP) kinases.

Formation of di & tri-phosphates

Nucleoside monophosphate (AMP, GMP)

- ATP
- NMP kinase
- ADP

- Nucleoside diphosphate (ADP, GDP)

- ATP
- NDP kinase
- ADP

- Nucleoside triphosphate (ATP, GTP)

Inhibitors of purine synthesis

- Folic acid (THF) is essential for the synthesis of purine nucleotides.
- Sulfonamides are the structural analogs of paraaminobenzoic acid (PABA).
- These sulfa drugs can inhibit the synthesis of folic acid by microorganisms.

- This indirectly reduces the synthesis of purines & nucleic acids (DNA & RNA).
- The structural analogs of folic acid (e.g. methotrexate), used to control cancer.
- They inhibit the synthesis of purine nucleotides & nucleic acids.
- These inhibitors also affect the proliferation of normally growing cells.

- **Azaserine (diazo acetyl-L-Serine) is a glutamine antagonist & inhibits reactions involving glutamine.**
- **Other synthetic nucleotide analogues used as anticancer agents are 6-thio guanine & 8-aza guanine.**

Salvage pathway for purines

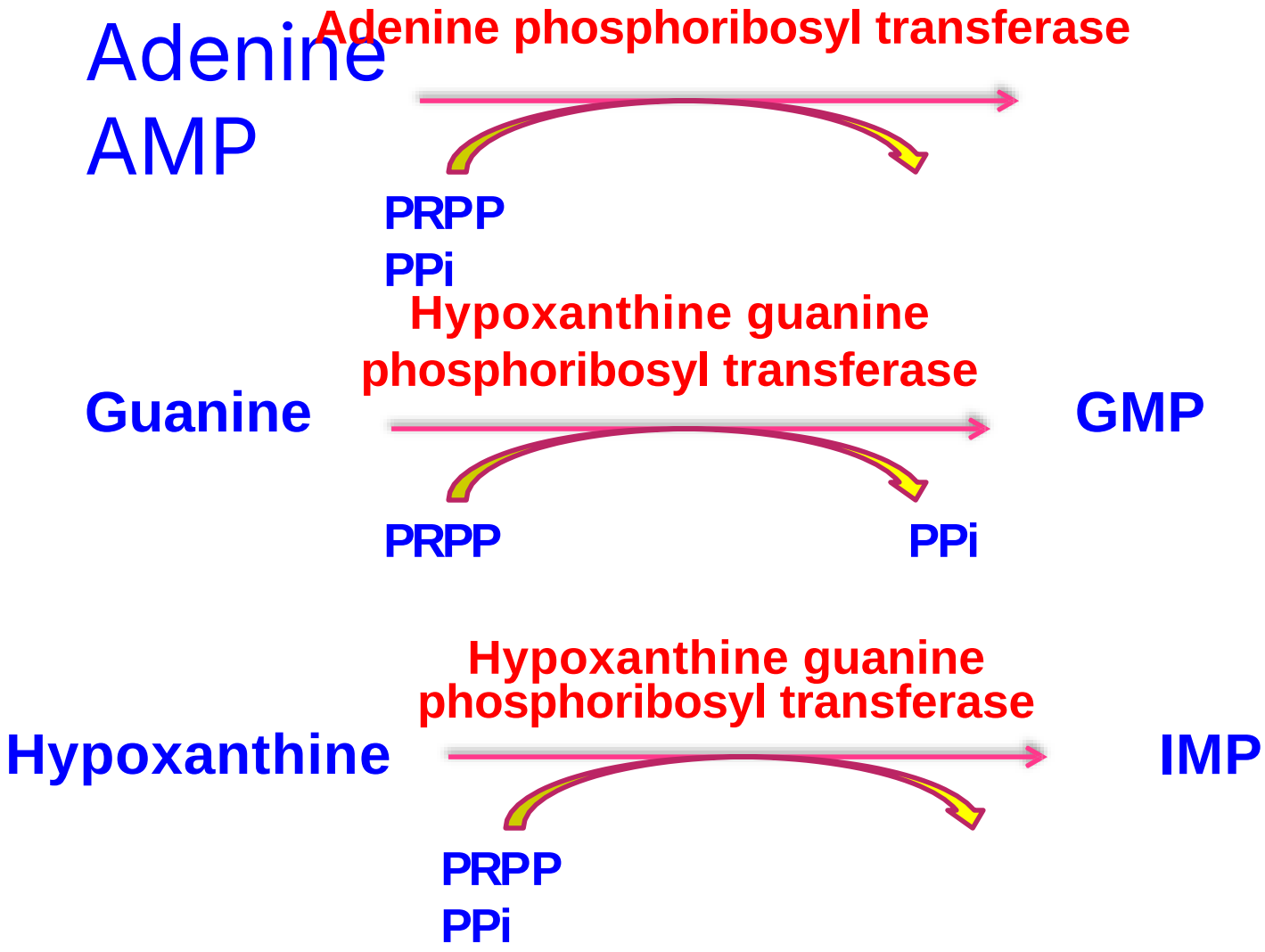
- This pathway ensures the recycling of purines formed by degradation of nucleotides.
- Nucleosides & deoxy-nucleosides can also be salvaged.
- The purines can be directly converted to the corresponding nucleotides & this process is known as ‘salvage pathway’.

- PRPP is the starting material in this pathway.
- It is also a substrate for de novo synthesis.
- The free purines are salvaged by two different enzymes.

1. Adenine phospho ribosyl transferase (APRTase).
2. Hypoxanthine guanine phosphoribosyl transferase (HGPRTase).

- **Adenine phosphoribosyl transferase catalyses the formation of AMP from adenine.**
- **Hypoxanthine-guanine phosphoribosyl transferase (HGPRT) converts guanine & hypoxanthine to GMP & IMP.**
- **Phosphoribosyl pyrophosphate (PRPP) is the donor of ribose 5-phosphate in the salvage pathway.**

Salvage pathway for purines



- The salvage pathway is particularly important in certain tissues such as erythrocytes & brain where de novo (a new) synthesis of purine nucleotides is not operative.
- A defect in the enzyme HGPRT causes Lesch

  - Nyhan syndrome.

Regulation of purine nucleotide synthesis

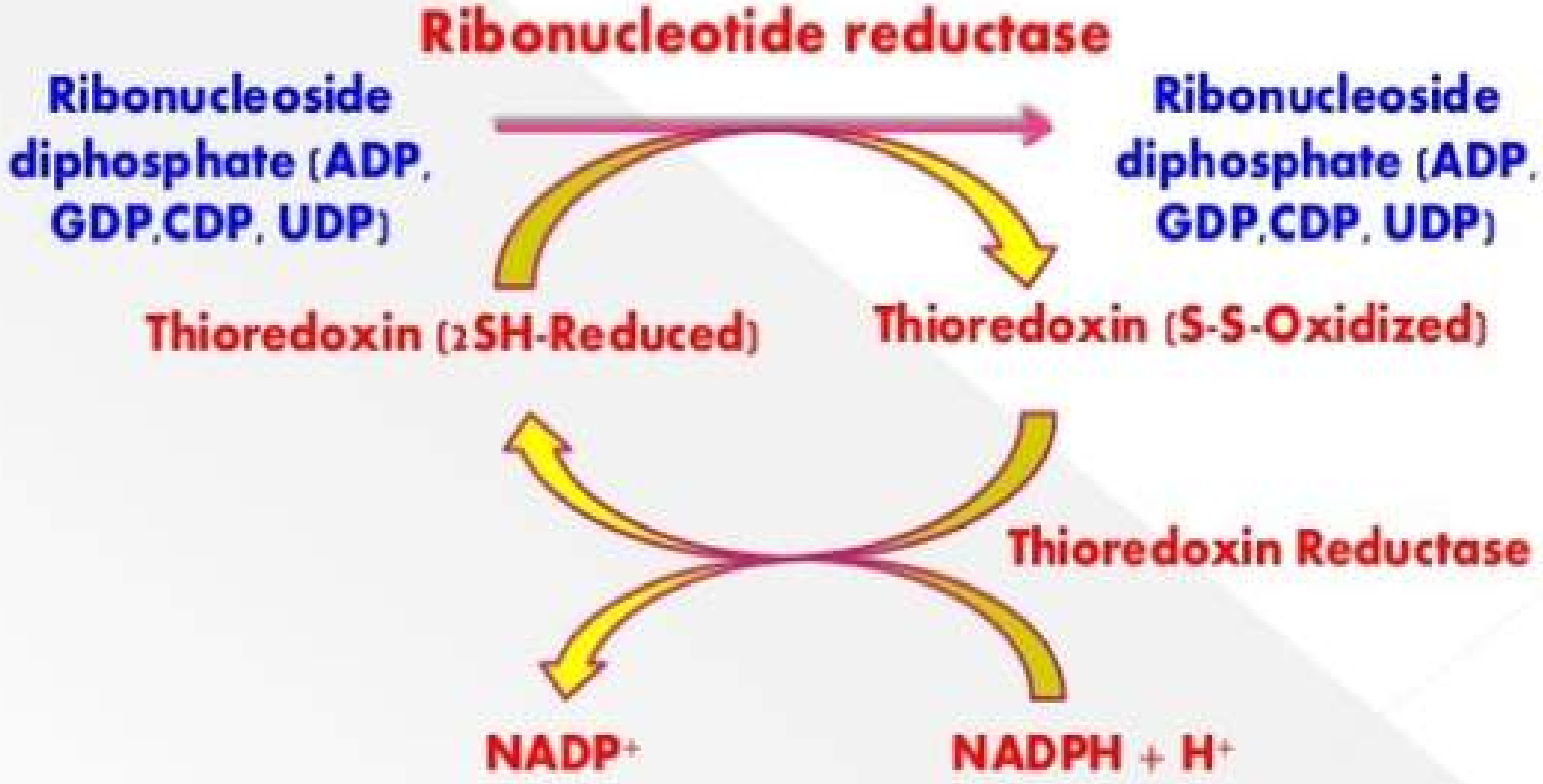
- The intracellular concentration of PRPP regulates purine synthesis.
- This is dependent on the availability of ribose 5-phosphate & the PRPP synthetase.
- PRPP glutamyl amidotransferase is controlled by a feedback mechanism by purine nucleotides.

- If AMP & GMP are available in adequate amounts, their synthesis is turned off at the amidotransferase reaction.
- Another important stage of regulation is in the conversion of IMP to AMP & GMP.
- AMP inhibits adenylsuccinate synthetase while GMP inhibits IMP dehydrogenase.
- AMP & GMP control their respective synthesis from IMP by a feedback mechanism.

Conversion of ribonucleotides to deoxyribonucleotides

- The synthesis of purine & pyrimidine deoxyribonucleotides occurs from ribonucleotides by a reduction at the C2 of ribose moiety.
- This reaction is catalyzed by a multisubunit (two B1 & two B2 subunits) enzyme, ribonucleotide reductase.

Formation of deoxyribonucleotides from ribonucleotides



- **Supply of reducing equivalents:**
- **The enzyme ribonucleotide reductase itself provides the hydrogen atoms needed for reduction from its sulfhydryl groups.**
- **The reducing equivalents, in turn, are supplied by thioredoxin, a monomeric protein with two cysteine residues.**

- **NADPH-dependent thioredoxin reductase converts the oxidized thioredoxin to reduced form which can be recycled again & again.**
- **Thioredoxin thus serves as a protein cofactor in an enzymatic reaction.**

- **Regulation of deoxyribonucleotide synthesis:**
- **Deoxyribonucleotides are mostly required for the synthesis of DNA.**
- **The enzyme ribonucleotide reductase maintains the adequate supply of deoxyribonucleotides.**
- **Ribonucleotide reductase is a complex enzyme with multiple sites (active site & allosteric sites) that control the formation of deoxyribonucleotides.**

References

- Lippincott
- **Chatterjea**